



Unpacking the process of conceptual leaping in the conduct of literature reviews

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ABSTRACT

Literature reviews serve diverse purposes, including description, understanding, explanation, and testing. Traditionally – before online databases, full-text search availability, and AI-based search tools – identifying relevant sources might have been considered a valuable contribution. However, top-tier information systems (IS) journals now demand more than descriptive reviews; they require authors to move beyond summarizing existing knowledge toward proposing innovative research directions, important research questions, new concepts, and interesting linkages among concepts. Despite adhering to rigorous methodological guidelines, many authors struggle to make conceptual leaps, that is, to elevate their literature reviews beyond description, to achieve a profound understanding, to provide explanations, or to develop a model. Authors may mistakenly prioritize hard work – like thorough literature search, analysis, and organization – over hard thinking, which is crucial for advancing theoretical contributions. With this in mind, I adopt the view that the literature is indeed qualitative data. I suggest that approaches that help make conceptual leaps in qualitative research can benefit literature review authors searching for inconsistencies in the extant literature and developing new questions, concepts, and linkages. Drawing upon qualitative research (Klag, M., and Langley, A., 2013. Approaching the conceptual leap in qualitative research. *International Journal of Management Reviews*. 15 (2), 149–166.), I unpack the process of conceptual leaping in the conduct of literature reviews. This process involves navigating dialectic tensions between *knowing* and *not knowing*, *engagement* and *detachment*, *deliberation* and *serendipity*, and *self-expression* and *social connection*. Effectively managing these tensions can help authors increase the impact and innovativeness of their literature reviews.

Introduction

When reviewing extant literature, authors may aim at different goals: description, understanding, explanation, or testing (Rowe, 2014). Descriptive reviews classify, organize, map, and report on the literature related to a topic or domain (e.g., Schlagenhauser and Amberg, 2015). Reviews aimed at understanding are often problem-focused, and their authors engage in better defining a phenomenon, a problem, or even a knowledge domain—often proposing a conceptual framework as an essential outcome (e.g., Vial, 2019). Explanatory reviews aim to elucidate why, how, and when things happen; their most common outcomes are models or theories (e.g., Oehlhorn et al., 2020). As for reviews for testing, they incorporate the results of extant empirical studies into a model that they test to determine the overall robustness of results from prior research (e.g., Schermann et al., 2016). Notwithstanding the genre they choose, literature review authors share a common aspiration: to make an impactful contribution to knowledge development in a domain of study.

Several decades ago, before large databases, full-text search availability, and AI-based tools for literature search, the mere identification of sources on a given topic might have been considered a contribution in its own right. Nowadays, however, descriptive

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reviews of the abovementioned genre can hardly find their way to publication in top-tier information systems (IS) journals. For instance, the *MIS Quarterly* Review department called for manuscripts that, in addition to presenting relevant prior literature in IS and related areas, also developed a model to guide future research, along with propositions justified by theoretical explanations, past empirical findings, and practical examples (Webster and Watson, 2002; Rivard, 2014). Other IS journals make similar calls for contributions, emphasizing understanding and explanation. As Editor-in-Chief of the *European Journal of Information Systems*, Rowe states that although descriptive reviews could play an important role when they pertain to emerging topics, “[t]o be worthy of appearing in top journals, there is a need to go beyond the descriptive literature type unless it bears on methodological and epistemological issues” (Rowe, 2014: 251). The *Journal of the Association for Information Systems*¹ website informs potential authors of literature review manuscripts that they are expected “to make a solid theoretical contribution.” The website stresses that its review and theory development section “will not consider papers that seek only to synthesize, summarize, or map a mature or emergent body of knowledge (e.g., narrative, descriptive, and scoping reviews) or reviews aimed at theory testing (e.g., meta-analysis)” (as described in Paré et al., 2015). Likewise, the editorial teams of the *Journal of Strategic Information Systems*’ annual literature review issue call for “high-impact scholarly surveys of important research” and invite authors to contribute manuscripts that “convey important implications for future research.”² They recently emphasized the role of the authors as the primary review instrument, notwithstanding the availability of powerful AI tools. They state that “[t]he key to making a compelling contribution is active engagement with literature searches, developing an in-depth understanding of the topic, and leveraging creativity and in-depth understanding to offer novel and valuable research directions in which generative AI tools can only serve as a supplement” (Pan et al., 2023).

However, my experience as a senior editor made me realize that many authors struggle to develop their literature reviews beyond the description level. Indeed, most review manuscripts I have handled reported faithfully following rigorous methodological recommendations, such as documenting the search terms and the databases used, the rules applied to select and reject sources, the period covered by the search, and so on (Paré et al., 2015). They also provided results of detailed analysis and classification of central concepts, underlying theories, research methods used, and main findings. While transparency regarding methods is essential for distinguishing scientific contributions from the layperson’s opinion (Ngwenyama and Rowe, 2024), many authors fail to make the conceptual leaps (Klag and Langley, 2013) that would take them from description to understanding, explaining, or developing a model for testing. Why is it so? Perhaps the bluntest and most honest explanation was provided by a reviewer who apologized to the authors for recommending a reject decision on a manuscript for which I was senior editor:

I am going to take a risk, and state what I believe to be the underlying reason, although there really isn't solid evidence to support this. I believe the authors have confused hard work with hard thinking. There's a lot of reading, and a lot of attempted organizing in this paper. But a theoretical paper needs well-developed new ideas. These might be triggered by reading but are more likely to be triggered by a cup of coffee in a relaxed environment.

The recommendation to rely on enjoying a cup of coffee in a relaxed environment that would favor the occurrence of conceptual leaps is indeed sound. I refer to this activity as a respite, essential in theory-building (Rivard, 2021). In brief, a respite is a form of combinatory play (Stevens, 2014) wherein an individual enters a state of playful manipulation of ideas, words, images, or objects, experiments with hypotheses, and explores various outcomes. Respite extends to instances where one does not actively focus on a problem to solve but allows one’s mind to wander during unrelated activities like running or walking the dog.

Although I genuinely believe in the power of respite to help authors make conceptual leaps between their well-read and well-analyzed literature and a deep understanding or rich explanation of the phenomenon they study, it may not suffice. In their editorial for the *JSIS* literature review issue, for instance, Pan et al. (2023) invite authors to “use their imagination and creativity in such a situation to connect seemingly unrelated ideas, unite different concepts, and produce fresh insights while building on existing literature across domains.” Following in those steps, I adopt the view of the literature – “the flux of papers, books, presentations, working papers, and so on representing the collective knowledge of the intellectual community” (Shepherd and Sutcliffe, 2011: 362) – as qualitative data, which implies that the work of reviewing the literature is akin to analyzing qualitative data. This view concurs with that of Wolfswinkel et al. (2013), who describe how grounded theory methods can assist the conduct of literature reviews, and of Boell and Cecez-Kecmanovic (2014), who propose a hermeneutic framework that portrays reviewing the literature as an iterative process of developing understanding. Accordingly, I surmise that approaches that facilitate the emergence of conceptual leaps in qualitative research are likely helpful for authors of literature reviews in their search for inconsistencies or shortcomings in the extant literature and their development of new themes, concepts, linkages, and dependencies among concepts.

Based on this view of the literature as qualitative data, I borrow from Klag and Langley’s (2013) notion of conceptual leaping in qualitative research and freely adapt it to the context of literature reviews. In the following paragraphs, after briefly defining conceptual leaps, I report on my understanding and my adaptation of Klag and Langley’s conceptual leaping process, which comprises four implicit dialectic tensions between *knowing* and *not knowing*, *engagement* and *detachment*, *deliberation* and *serendipity*, and *self-expression* and *social connection*. I conclude with recommendations on enacting this dialectical process.

Conceptual leaps and conceptual leaping

In the context of qualitative research, Klag and Langley (2013) use the term conceptual leap—adopted here—to refer to an

¹ <https://aisel.aisnet.org/jais/authorinfo.html>, consulted February 8th, 2024.

² Galliers, R.D., Jarvenpaa, S. L., Rivard, S., *The Journal of Strategic Information Systems* original call for proposals for review articles, circa 2018.

Table 1
Areas of tension (Source: Klag and Langley, 2013: 152).

Four areas of tension between states of mind	
<i>Knowing</i> : Leveraging experience and expertise	<i>Not knowing</i> : Embracing doubt, leveraging naïveté
<i>Engagement</i> : Exercising data to seed ideas	<i>Detachment</i> : Incubation to grow ideas
<i>Deliberation</i> : Using heuristics to stimulate variety	<i>Serendipity</i> : Leveraging chance to inject novelty
<i>Self-expression</i> : Leveraging subjectivity	<i>Social connection</i> : Leveraging embeddedness

intellectual jump that “generates abstract theoretical ideas from empirical data” (p. 149). The authors refer to theoretical ideas to encompass various conceptual products, including but not restricted to concepts, conceptual insights, conceptual distinctions, substantive theories, and formal theories. I suggest that the authors of a literature review must make similar conceptual leaps between the swamp of articles, books, conference proceedings, and working papers they gather and the conceptual products that premier information systems journals call for when they invite literature reviews: important issues, new research questions, understandings, or explanations via concepts, frameworks, models, propositions, or theories.

Submitting that “the path towards integrated insight is often punctuated by multiple [conceptual leaps]” (p. 151), some small, some more significant, Klag and Langley propose the notion of a conceptual leaping process. Indeed, it is seldom that a new concept—let alone a new theory—will instantly emerge from the researcher experiencing “sparks of inspiration [...] during trancelike periods of intensive writing” (Rivard, 2021: 317). The idea that several conceptual leaps “emerge over time through connections between data and ideas suggests broadening the focus beyond the conceptual leap as a singular magical event to consider conceptual leaping as a process” (Klag and Langley, 2013: 151). Klag and Langley further depict conceptual leaping as dialectical, based on their observation that literature about the occurrence of conceptual leaps comprises multiple instances of contradictions or oppositions, be it within a single author’s discourse or among different authors’ standpoints. For example, some authors posit that a conceptual leap is more likely to occur when the researcher alternates between states of mind, such as Weick’s representation of theory construction as “*disciplined imagination*” put into action or Locke’s (2007) emphasis on the importance of adopting a “*dual-thinking mode*” when conducting grounded theory research. Other contradictions emerge among different authors’ recommendations. Here, Klag and Langley refer to some authors recommending the purposeful use of ‘methods of discovery’ (e.g., Abbott, 2004) while others advocate for relying on serendipity (e.g., Fine and Deegan, 1996); some advise researchers to stay close to their data (Glaser, 1978), and others stress the importance of incubation [that can occur during a walk in the park] (Mellor, 2001: 473). Analysis of the literature led Klag and Langley to extract four dialectic tensions that, when mobilized alternately and brought into synthesis “by building on both poles, but ensuring that neither dominates for too long” (p. 161), constitute the process of conceptual leaping. They propose dialectic tensions between *deliberation* and *serendipity*, *engagement* and *detachment*, *knowing* and *not knowing*, and *self-expression* and *social connection* (see Table 1).

Importantly, Klag and Langley’s depiction of the conceptual leaping process is not prescriptive. Indeed, they present it as “a ‘do-it-yourself’ process of ‘cobbling together’ that one undertakes with the tools at hand” (p. 162). While I agree with this viewpoint, my experience as an author, supervisor, and editor leads me to suggest that in the context of a literature review, conceptual leaping almost naturally commences with the dialectic tension of *knowing* and *not knowing*. Furthermore, I find it helpful to characterize the areas of tension—in the context of a literature review—as opposite, albeit complementary, *states of mind* between which an author can purposefully, consciously, and recursively alternate.

Knowing vs. not Knowing

In their basic definitions, *knowing* refers to having, possessing, or showing knowledge about something; *not knowing* refers to not being informed or being ignorant about something. In the context of a literature review, the expert researcher in a domain is the archetype of *knowing*, and the researcher who is new to the field is that of *not knowing*. However, engaging in a conceptual leaping process implies that notwithstanding their level of expertise or erudition in a research domain, authors purposefully adopt each state of mind alternately and iteratively while they conduct a literature review.

Undertaking a literature review as a field expert gives the researcher a good head start. They already know who the primary authors are, understand the key concepts, are cognizant of the important theories, and are aware of the critical issues. However, this familiarity can be a source of bias as it may prevent the author from making discoveries in the literature—and even prevent them from reading it attentively! “Consequently, there is less of an opportunity for the research problems and solutions to be generated from the Literature itself” (Shepherd and Sutcliffe, 2011: 369). Therefore, the field expert may want to temporarily “adopt an approach of willful suspension of belief” (as Gioia refers to in a qualitative research context—Gioia et al., 2011: 292), put themselves in a mindset of *not knowing*, and read and interpret the literature with fresh eyes. To reinforce this, Wolfswinkel et al. (2014) even recommend, once the relevant literature has been identified, randomly picking each paper to be read instead of reading them in any predetermined order—e.g., chronological, alphabetical, or sub-areas. Whether the researcher is an expert or a novice—the latter naturally entering the literature in a *not-knowing* state of mind—they then engage with the literature in a way that allows their reading to generate an overall understanding of the phenomenon they study. This understanding then guides their focus toward themes, concepts, and theories that constitute an opportunity for additional attention when they perceive a “tension, opposition, and/or contradiction” in this literature (Shepherd and Sutcliffe, 2011: 368).

In contrast, when adopting a state of mind of *knowing*, the researcher leverages their experience and expertise to nurture the generation of conceptual leaps (Klag and Langley, 2013). According to Klag and Langley, there are two ways by which experience and

expertise can assist the conceptual leaping process: the researcher's accumulated substantive and theoretical knowledge and the experience of conceptual leaping acquired through practice. When the literature constitutes the data to be analyzed, [Shepherd and Sutcliffe \(2011\)](#) refer to the researcher's knowledge that results from their "own interests, competencies and expertise, body of work (e.g., publications), and experiences (e.g., conference presentations or conferences attended, articles he or she has reviewed for journals, doctoral courses taken and taught)" (p. 369). While overreliance on this knowledge may diminish the opportunity for concepts, gaps, explanations, or tensions to emerge from the literature, authors can nevertheless successfully mobilize their knowledge to assess the discoveries made when they were in a *not-knowing* state of mind, at times to resolve tensions, at other times to explore those tensions further and to direct the attention towards discoveries. Notably, *knowing* is deemed critical to reaching "appropriate and defensible meanings for the epistemic context [of the review]," which "requires the scientist to be socialized in the relevant epistemic community and deeply engaged in the emergent conversation on the subject matter ([Toulmin, 1985](#))" ([Ngwenyama and Rowe, 2024](#): 124). Accumulated knowledge, however, is not the prerogative of the sole field expert. Even a novice researcher has acquired knowledge and has been socialized through former training, courses, manuscripts reviewed, conference presentations made or attended, and everyday life experiences. This knowledge, too, can be successfully leveraged in analyzing, understanding, and interpreting the literature.

Knowing, in terms of the researcher's experience of conceptual leaping acquired through practice, is likely to accelerate or facilitate the emergence of conceptual leaps. Deemed to be tacit knowledge, such an experience appears to be better transmitted via apprenticeship. Although apprenticeship is likely to allow a novice researcher to acquire experience in conceptual leaping, it "may undermine the potential of the heuristic solutions suggested [in the discussion on *elaboration* below], as well as the hope that exercising data by the book will work any magic without experience under the tutelage of a master" ([Klag and Langley, 2013](#): 158).

Engagement vs. Detachment

Whether in analyzing the literature or qualitative data, *engagement* refers to the conscious immersion into the data, 'exercising' them to 'seed ideas' ([Klag and Langley, 2013](#): 155). The opposite state of mind is that of *detachment*, a mode of 'incubation' defined by [Wallas \(1926\)](#) as a "temporary shift away from an unsolved problem that allows a solution to emerge seemingly as if from no additional effort" ([Sio and Ormerod, 2009](#): 94).

When entering an *engagement* state of mind, the researchers immerse themselves in the documents they have accumulated. Here, authors are expected to "organize, compare, collate, summarize, aggregate or interpret the information previously extracted to suggest a new contribution to knowledge" ([Templier and Paré, 2015](#): 116). Many recommendations exist regarding analyzing the literature in an *engagement* mode. For instance, [Wolfswinkel et al. \(2014\)](#) suggest adopting a grounded theory method of analysis wherein the researcher reads each document and highlights the relevant text excerpts. Researchers should then use the grounded theory coding approach: "[O]pen coding is the analytical process of generating higher-abstraction level type categories from sets of concepts/variables. Axial coding is further developing categories and relating them to their possible sub-categories. With selective coding, the categories are integrated and refined" (p. 50). Creating tables that help organize the literature is another analysis approach that can complement grounded-theory coding methods. In this context, [Webster and Watson \(2002\)](#) refer to two types of tables: author-centric and concept-centric tables (see [Table 2](#)). Although concept-centric tables are more amenable to developing an understanding of the literature, my experience is that building author-centric tables is easier and more efficient when engaging in the literature in a *not-knowing* state of mind. Such a table can constitute a document base of the literature, and its building can be pursued even when a manuscript is revised. Conscious *engagement* with the literature is likely to imply the analysis of the author-centric tables, for example, examining whether or how some concepts were studied together, whether relationships have been hypothesized or found in extant research, and so on ([Holton and Lowe, 2007](#)). Then, the researcher can transform an author-centric table into a concept-centric table.³

Although necessary, the activities performed in an *engagement* state of mind often fail to yield a conceptual leap. As the reviewer mentioned above commented, well-developed novel ideas "might be triggered by reading but are more likely to be triggered by a cup of coffee in a relaxed environment." This reviewer referred to a *detachment* state of mind ([Klag and Langley, 2013](#)). Examples abound as to how getting away from one's data or tentative explanations can be productive. For instance:

In his struggles with extremely complicated mathematics that led to the general theory of relativity of 1915, Einstein often turned for inspiration to the simple beauty of Mozart's music [...] "Whenever he felt that he had come to the end of the road or into a difficult

Table 2
Concept-centric and author-centric tables ([Webster and Watson, 2002](#): xvii).

Approaches to Literature Reviews	
Concept-centric	Author-centric
Concept X ... [author A, author B, ...]	Author A ... concept X, concept Y, ...
Concept Y ... [author A, author C, ...]	Author B ... concept X, concept W, ...

³ For instance, in [Kappos and Rivard \(2008\)](#), the Appendix of the article is the author-centric table that resulted from a first level of analysis of the 56 articles of their document base. The article's table 3 is a concept-centric table that resulted from additional rounds of analysis, navigating between engagement and detachment.

situation in his work, he would take refuge in music," recalled his older son, Hans Albert. "That would usually resolve all his difficulties." (Miller, 2006).

Detachment pertains to the notion of combinatory play, which "describes the conscious and unconscious cognitive playful manipulation of two or more ideas, feelings, sensory experiences, images, sounds, words, or objects. In combinatory play, players experiment with hypotheses, they play with possible outcomes, and they adjust to unexpected results and even "failures" (Stevens, 2014: 99). The spiral model of theory building I have proposed (Rivard, 2021) refers to combinatory play as "respite" from the explanation-development effort that leaves the researcher's mind exhausted:

[D]uring respite, "the conscious mind remains busy with other tasks, but the unconscious mind keeps working on the problem, combining or playing with ideas in ways rational thought might inhibit" (Stevens, 2014: 102). According to Stevens, this is when the "illumination" of the explanation can emerge. Although the outcome of combinatory play is not always an illumination in the sense of a "eureka" moment, it can nevertheless lead theorists to a more interesting explanation and a well-organized, clear, and interesting presentation (Rivard, 2021: 325).

However, neither *engagement* with the literature nor *detachment* suffice to generate a conceptual leap. Alternating between the two poles and reaching synthesis will potentially be more fruitful.

Deliberation vs. Serendipity

Even though there exist similarities between *deliberation* vs. *serendipity* and *engagement* vs. *detachment*, I agree with Klag and Langley that the two sets of states of mind differ. While *engagement* implies complete immersion in reading and analyzing one's data (here, the literature), *deliberation* refers to consciously applying heuristics to enrich the understanding that has emerged during an *engagement* vs. *detachment* synthesis. While *detachment* refers to letting the mind wander, *serendipity* refers to "the discovery by chance of valid but unsought elements by a mind that is theoretically prepared" (Manuscalo, 1998).

My understanding is that the activities conducted when being in a *deliberation* state of mind during a literature review are at a higher level of abstraction than those undertaken in an *engagement* state of mind. During *deliberation*, it is not the literature itself that is analyzed but the understanding of this literature the researcher has already developed: the concepts that have emerged, the relationships that have been observed in the literature, or the gaps that have appeared.

Various approaches can assist researchers when they enter *deliberation*. For instance, Jaccard and Jacoby (2020) propose 27 heuristics that can foster researchers' creativity by making them think about the phenomena they study in a way that might provide new insight. Although all twenty-seven are not likely to be helpful in a literature review context, many may help. For instance, Jaccard and Jacoby propose "to make the opposite assumption than that existing in a body of works." This heuristic is akin to the problematization approach that Alvesson and Sandberg (2011) introduce. In this approach, researchers analyze the literature to uncover widely held assumptions, challenge those assumptions, and propose alternate assumptions that can enrich the understanding of a phenomenon and inspire future research. Other heuristics, such as conducting a thought experiment, can be used to assess the soundness of alternative assumptions (Jaccard and Jacoby, 2020). Another approach is to compare the assumptions of different bodies of literature about a phenomenon (e.g., Moeini and Rivard, 2019). To do this, the researcher has to have accomplished at least one iteration of *engagement* and *detachment* and have gained a good understanding of the literature. If, in doing so, the researcher has observed that this literature comprises different bodies anchored in different, sometimes opposed assumptions, they can conduct a dialectical interrogation of these bodies to articulate their key assumption-level tensions. Many other heuristics proposed by Jaccard and Jacoby (2020)—e.g., reframing a problem in terms of the opposite, applying the continual why and what, using counterfactuals, pushing an existing finding to the extremes, role-playing, or applying deviant case analysis—can be mobilized when working in a *deliberation* state of mind. Although authors sometimes naturally apply one or a few heuristics, my experience is that purposefully selecting a few and mindfully applying them often catalyzes one's creative thinking.

While *deliberation* implies a state of mind under which researchers purposefully "use heuristics to stimulate variety," a *serendipity* state of mind leverages "chance to inject novelty" (Klag and Langley, 2013: 154). The first use of the term *serendipity* is attributed to Horace Walpole, who, in 1754, referred to it as "making discoveries by accident and sagacity, of things which one is not in quest of."⁴ *Serendipity* has often been attributed an important role in discoveries, from Newton's law of gravity and Fleming's discovery of penicillin to Roentgen's "invention" of X-rays. *Serendipity* is often referred to as the "discovery by chance" of something. Although chance plays a key role here, sagacity—i.e., the acuteness of mental discernment and soundness of judgment—as coined by Walpole, is an essential ingredient for the serendipitous incident to morph into an important discovery. Indeed, how many people have observed apples falling from trees without coming up with a law of gravity?

In social science research, *serendipity* often refers to "the discovery by chance or sagacity, of valid results which were not sought for" and which allows generating new hypotheses (Merton, 1945: 469). A serendipitous experience often refers to discovering unsought elements in the material—empirical observations or literature—that the researcher analyzes, such as unexpected behavior in one's data or intriguing contradictory results observed in different studies. This happened during the development of Kappos and Rivard's (2008) literature review on culture and information systems development and use. The conceptualization of culture adopted in 55 of the 56 articles the authors had selected belonged to either an integration or a differentiation perspective. One article was an outlier:

⁴ <https://www.merriam-webster.com/wordplay/happy-birthday-serendipity>.

Dubé and Robey (1999) adopted a conceptualization of culture (Martin, 1992), wherein three perspectives jointly exist within any collective—integration, differentiation, and fragmentation. Instead of considering this discovery a simple blip, Kappos and Rivard seized this incongruity, engaged with Martin's theory of culture, and adopted her three-level conceptualization. This decision led them to re-enter the literature and go one more time through the cycles of *knowing-not knowing*, *engagement-detachment*, and *deliberation-serendipity*. I believe the model that ultimately emerged from seizing this serendipitous discovery was more interesting than it would have been had the authors ignored the blip!

Although the above-described serendipitous discoveries emerged from the literature the researchers were reviewing, my experience is that they can also occur during one's activities outside the literature review—more like Archimedes' bath or Newton's apple. I can remember, for instance, a lunchtime discussion with colleagues where someone mentioned a specific contextual variable in the research he was conducting. This researcher was from a field of study not related to mine. But grabbing the contextual variable the colleague had referred to led me to (1) take a moment to come to a synthesis between this potential new contextual variable and the results of my *deliberation* efforts—here, exploring whether the contextual variable was relevant to my review and (2) go back to “my literature,” and re-enter a new cycle of *knowing-not knowing*, *engagement-detachment*, and *deliberation-serendipity*.

Self-expression vs. Social Connection

While descriptive literature reviews might call for “eliminating any subjectivity which may interfere with the accuracy of aggregated evidence or descriptive overviews” (Wagner et al., 2022: 214), I contend that reviews that aim at understanding and explaining require that authors enter a state of mind of *self-expression* that leverages subjectivity (Klag and Langley, 2013: 159) and allow them to exercise their creativity. In this line of thought, Shepherd and Sutcliffe (2011) build on the coherence theory of perception (Rensink, 2002) and posit that before a researcher develops a conceptual representation of the literature, they form a sensory representation, which is a subjective representation, “a *perception* of tensions, oppositions, and/or contradictions among divergent perspectives and explanations of the same phenomena in the Literature” (p. 366). Based on this sensory representation, the researcher builds the conceptual representation they will report in their manuscript.

Adopting a *self-expression* state of mind demands that the researcher goes beyond the essential activities of classifying, organizing, mapping, and reporting on the literature. This implies that authors should “avoid mechanizing the process” that would lead one to eschew “the creative processes that are essential to an effective review” (Leidner, 2018: 563). Indeed, literature reviews are portrayed as “inherently interpretive processes in which a reader engages in ever-expanding and deepening understanding of a relevant body of literature” (Boell and Ceez-Kecmanovic, 2014: 259). *Self-expression* may imply that authors commit to certain positions and develop strong arguments supporting their proposed new conceptualizations. For instance, in their review of the IS literature on participation—a widely acknowledged IS concept—Markus and Mao exercise their *self-expression* in qualifying the in-use concept of participation as an “old, tired concept for today's IS contexts” (Markus and Mao 2004, p. 514). However, the authors do not merely make this solid personal statement; they support it with an in-depth literature analysis and the proposition of a well-justified new conceptualization and new hypotheses. Indeed, their article:

[...] critically analyzes traditional IS participation theory and shows that it contains partial and conflicting explanations for participation's effects on system success. [The article shows how] these explanations leave important conceptual issues unresolved [...]. To address these gaps, [the article] outlines the key elements of a new theoretical framework, including a redefined concept of system success, an elaborated conceptualization of participants and other actors, a fine-grained characterization of participation activities, and a restatement of hypothesized causal links among the concepts (Markus and Mao, 2004: 514).

Notwithstanding the paramount role of *self-expression* in the conceptual leaping process, the insights that one generates when engaged in their work, “are unlikely to spread or develop unless they resonate with others” (Klag and Langley, 2013: 160). Accordingly, leveraging embeddedness through *social connections* is critical to conceptual leaping. Adopting a *social connection* state of mind can be as informal as discussing a topic with a colleague over coffee, meeting with one's supervisor, or asking for a friendly review. It can also be more explicit when authors present their work to conferences and “workshop” their manuscript to receive formal feedback. To benefit from a *social connection* state of mind, one ought to come prepared: one must be open and listen to comments and suggestions while at the same time not losing sight of one's aim. Indeed, it can sometimes be tempting to resist specific comments or suggestions because they might involve renouncing (Rivard, 2021) some element of a review that has required hard crafting. At other times, however, junior researchers can be swiftly steered from their original focus because someone in their audience who has listened to a 10-minute review paper presentation suggests a significant change of orientation. As observed by Klag and Langley: “as inspiring and generative as social connectedness may be during the process of conceptual leaping, it must also be noted that, in some cases, formal or informal dialogue with others may act as a constraint rather than an enabler” (p. 161).

Enacting the process of conceptual leaping

During discussions about conceptual leaping in conducting literature reviews, I was often asked what the most efficient way was of enacting the process. Should one navigate it in the order in which the states of mind are presented here or in another? My immediate response was that there is no single way, no template for action. I embrace Klag and Langley's view that the dialectic tensions presented here are tools available to the researcher for trying, tinkering, playing, and testing (p. 161). When challenged by my discussion partners to provide some guidance, nevertheless, the paramount role of mindfulness – i.e., “paying attention to internal and external present-moment events, [which] involves directing attention toward not only external events and phenomena but also internal events”

(Sutcliffe et al., 2016: 59) – came to my mind.

Accordingly, my first recommendation for a fruitful enactment of the conceptual leaping process is to be mindful of one's current state of mind, identify it, and purposely "flip" to its opposite after a while. For instance, after several hours of *engagement*, when the mind is caught in a detailed literature analysis, one should purposely detach from this work and let their mind wander. The literature suggests low cognitive demand tasks have a more substantial incubation effect than highly cognitive tasks or rest (Sio and Ormerod, 2009). Personal experience suggests that physical activities such as walking, running, or cycling have a strong incubation effect. Research indicates that even "simple distraction tasks, such as household chores, seem to allow time for creative incubation, thereby fostering creative insight" (Steindorf et al., 2021: 584). Similarly, after days of *self-expression*, one should consider the benefit of entering a *social connection* state of mind, whether discussing their work with a colleague or presenting it more formally.

My second recommendation related to mindfulness is to not rely on the sole power of one's thinking abilities to naturally guide one to engage in each of the four dialectic tensions but to alternate between them mindfully and purposefully. We often resist entering a new dialectic tension because doing so will likely challenge the work already accomplished. For instance, one might resist entering the *deliberation-serendipity* dialectic tension after being involved in an *engagement-detachment* tension or entering the *self-expression* vs. *social connection*. Indeed, I know from experience that submitting one's work to heuristics, accepting blips as serendipitous findings, or considering a colleague's suggestion during a workshop may challenge what has already been found. The perspective of having to recommence the entire *knowing-not-knowing*, *engagement-detachment*, *deliberation-serendipity*, and *self-expression-social connection* process, even with the hope of the emergence of a conceptual leap, can be daunting.⁵

My third recommendation draws on the idea that mindfulness not only pertains to the state of mind one has adopted but also to the awareness of the doubt, the contradictions, or the new ideas that may have emerged from alternating between states of mind. What concept, relationship, resemblance, or difference between study findings emerged during a period of *detachment* that followed several hours of *engagement*? How different was it from the observations made in a state of mind of *engagement*? What "blip" has been perceived in the literature or in one's reasoning while applying heuristics when being in a *deliberation* state of mind? Paying attention to the differences, conflictual observations, or divergent ideas that emerged from having alternated between two states of mind is essential to reaching synthesis, acknowledging whether a conceptual leap has occurred, and determining the best way forward.

I also embrace Klag and Langley's emphasis on the role of writing while navigating the conceptual leaping process: "[I]t is often in the very act of writing that productive conceptual ideas develop and crystallize" (p. 162). Not only is writing essential, but so is re-writing. Indeed, "an important aspect of creating good literature reviews is re-writing: they need to be re-written several times in order to form a better understanding and to better convey this understanding to readers" (Boell and Ceez-Kecmanovic, 2014). This resembles Rivard's emphasis on the writing activity while developing theory—either from qualitative data or from the literature: "Writing can take the form of jotting down ideas, documenting findings or hunches, sketching a model, and putting into words the actual theory piece. No matter its form, writing is essential throughout the theory-building process to capture the explanation as it develops" (Rivard, 2021: 318).

Concluding remarks

My understanding of conceptual leaping in the context of a literature review is that the process is highly iterative in addition to being dialectical. I suggest that the states of mind presented here are located on dialectical circles and that researchers navigate from one state of mind to another on a given circle. Following Van de Ven and Poole (1995), I suggest that the outcomes of the researcher's work when they alternately adopt two states of mind somewhat confront one another. The result of this confrontation is different from the individual outcome of having adopted each opposite state of mind. According to Van de Ven and Poole (1995) the result is "either of a new entity that is different from the previous two, or (in degenerate cases) the defeat of one entity by the other, or a stalemate among the entities" (p. 526). Transposed in a literature review context, the first possible result is a new concept, idea, gap, explanation, or understanding, different from the previous two—each having emerged from the researcher spending some time in each state of mind. The second possible result is the defeat of one outcome by the other—the inability of the researcher to come to a synthesis between the understanding that emerged from adopting either state of mind and the dominance of one outcome. The third possible result is a stalemate among the outcomes. In the context of a literature review, the latter often occurs when the researcher's "hard thinking" efforts result in a dead-end. In this case, resilience is critical. The researcher might have to re-commence their work (Rivard, 2021) and re-enter another dialectical circle. Here, I will conclude with the following wise words:

According to the Oxford English Dictionary (2008), to "review" means to "view, inspect, or examine a second time or again." [...] Sometimes many times (Michailova, 2023: 363).

Credit authorship contribution statement

Suzanne Rivard: Conceptualization, Formal analysis, Methodology, Writing – original draft, Writing – review & editing.

⁵ I do not imply here that entering a new dialectic tension always means that it will result in the need to start anew the whole conceptual leaping process. Although this is sometimes the case, it also happens that only relatively minor retrofitting is required.

Declaration of competing interest

The author declare that she has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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